

Lesson 04 · Powers, logarithms and exponential growth

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Without using your calculator, evaluate 8^2 .

☐ A 16

☐ B 64

☐ C 10

☐ D 256

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Q2 · Rank 1 / Foundation

Without using your calculator, evaluate 3^{-1} .

☐ A -3

☐ B 3

☐ C $\frac{1}{3}$

☐ D $-\frac{1}{3}$

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Q3 · Rank 1 / Foundation

Without using your calculator, evaluate $36^{1/2}$.

☐ A 18

☐ B 72

☐ C 1296

☐ D 6

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Q4 · Rank 2 / Core

Use the rules of indices to simplify $a^3 \times a^8$.

☐ A a^{11}

☐ B a^{24}

☐ C $2a^{11}$

☐ D a^5

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Q5 · Rank 2 / Core

Use the rules of logs to expand $\ln(xy)$. Assume $x > 0$ and $y > 0$.

☐ A $\ln x \ln y$

☐ B $\ln x + \ln y$

☐ C $\ln x - \ln y$

☐ D $x + y$

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Q6 · Rank 2 / Core

Solve $e^x = 5.9$. Round your answer to two decimal places.

☐ A 5.90

☐ B 0.59

☐ C 1.77

☐ D -1.77

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Q7 · Rank 3 / Application

The number of items, N , produced each day by an assembly-line worker, t days after an initial training period, is modelled by $N = 100 - 100e^{-0.4t}$. Calculate the number of items produced daily 1 day after the training period. Choose the value to two decimal places.

☐ A 67.03☐ B 100☐ C 40.00☐ D 32.97

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Q8 · Rank 3 / Application

For the worker model $N = 100 - 100e^{-0.4t}$, what is the worker's daily production in the long run?

☐ A 100 items/day☐ B 0 items/day☐ C 40 items/day☐ D Unlimited growth

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Q9 · Rank 3 / Application

Solve $\ln x = 0$. Which x intercept does this give for $y = \ln x$?

☐ A $x = 0$

☐ B $x = 1$

☐ C $x = e$

☐ D $x = -1$

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Q10 · Rank 4 / Challenge · OPTIONAL

Future sales of two products A and B are given by $S_A = 5e^{0.01t}$ and $S_B = 2e^{0.02t}$. Find the time, t , when sales of the two products are the same.

☐ A $\ln(2.5)$

☐ B $100 \ln(0.4)$

☐ C $100 \ln(2.5)$

☐ D $50 \ln(2.5)$

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